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Assembly for an aircraft comprising a wing and an engine pylon for coupling a propulsion system to said wing

Abstract

An assembly comprising a wing with a lower surface panel and an upper surface panel and an engine pylon comprising a primary structure fixed to the wing with a fixing base. The fixing base comprises fourth fixing yokes while the primary structure bears fourth fittings, the fourth fixing yokes being secured to the fourth fittings with fourth rods.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) This application claims the benefit of French Patent Application Number 2305084 filed on May 23, 2023, the entire disclosure of which is incorporated herein by way of reference.

FIELD OF THE INVENTION

(2) The present invention relates to an assembly for an aircraft which comprises a wing and an engine pylon for coupling a propulsion system to the wing, as well as an aircraft comprising a propulsion system and such an assembly for coupling the propulsion system to the wing.

BACKGROUND OF THE INVENTION

(3) Usually, for an aircraft, a propulsion assembly comprises, for example, a turbojet engine which is fixed to a wing of the aircraft using an engine pylon. The engine pylon is generally composed of a primary structure formed by a box composed of a top spar, a bottom spar and two lateral panels linking the two spars and internal ribs distributed along the box.

(4) The turbojet engine is fixed under the engine pylon by means of engine attachments which conventionally comprise, at the front, a front engine attachment, at the rear, a rear engine attachment, and, between the front and rear engine attachments, a thrust force absorption assembly

comprising absorption rods fixed between the turbojet engine and the primary structure of the pylon to absorb the thrust forces generated by the turbojet engine.
(5) The engine pylon is also fixed to the structure of the wing using a fixing base. More specifically, the fixing base has a substantially inverted U form and is intended to receive the rear end of the box. The base also bears the attachments of the engine pylon to secure the latter to the structure of the wing using fittings through which the forces originating from the turbojet engine pass to the structure of the wing.
(6) Although the current installations are satisfactory, the size of turbojet engines is increasing but it is not preferable to modify the dimensions of the landing gear so as to modify the height of the wing with respect to the ground for reasons of weight and bulk penalties. Now, the box has relatively significant dimensions which make the positioning of the turbojet engine under the wing difficult.
(7) It is therefore necessary to provide a solution that ensures an optimal installation of the turbojet engines under the wings of the aircraft.
SUMMARY OF THE INVENTION
(8) One object of the present invention is to propose an assembly for an aircraft comprising a wing and an engine pylon for coupling a propulsion system to the wing and which comprises means for fixing to the wing which make it possible to reduce the bulk of the engine pylon while ensuring an optimized transfer of the forces to the structure of the wing.
(9) To this end an assembly for mounting, on an aircraft, a propulsion system having a vertical median plane is proposed, comprising: a wing with a front spar, an upper surface panel fixed in the top part of said front spar and a lower surface panel fixed in the bottom part of said front spar, in which the front spar bears a first fitting at the vertical median plane, in which the lower surface panel bears two starboard and port fittings at a bottom part of said front spar, and in which the lower surface bears a third fitting positioned at the rear of the second starboard and port fittings, an engine pylon comprising a primary structure, a shackle fixed by a fifth link to said third fitting of said lower surface, said fifth link having at least one degree of freedom in rotation, a first and a second rod, in which a first end of each first and second rod is fixed to the shackle respectively by a third and a fourth pivot link, a third rod, in which a first end is fixed to the first fitting by a seventh link, said seventh link having at least one degree of freedom in rotation, and a fixing base comprising a body secured to a rear part of the primary structure.
(10) Said body comprises two first starboard and port fixing yokes extending to the rear and in the top part on the sides of said fixing base, each of said first fixing yokes being fixed to one of said second fittings by a first pivot link.
(11) Said body also comprises two second starboard and port fixing yokes extending to the rear and in the bottom part on the sides of said fixing base, each of said two second fixing yokes being fixed by a second link to a second end respectively of the first and second rods. Said second link has at least one degree of freedom in rotation.
(12) Said body further comprises at least one central third fixing yoke extending to the rear and at the vertical median plane, said central third fixing yoke being fixed to a second end of said third rod by a sixth link, said sixth link having at least one degree of freedom in rotation.
(13) According to the invention, the assembly comprises a fourth starboard fitting and a fourth port fitting, each of said fourth fittings being fixed to the primary structure of the engine pylon. For each fourth fitting, the assembly comprises a fourth rod, a first end of which is fixed to said fourth fitting by an eighth pivot link.
(14) Furthermore, said body comprises two fourth starboard and port fixing yokes extending to the front and in the bottom part on the sides of said fixing base, each of said fourth fixing yokes being fixed to a second end of a fourth rod by a ninth pivot link.
(15) With such an engine pylon, the forces from the engine which are conveyed by the primary structure of the pylon are transmitted directly to the upper surface and lower surface panels via the

fixing base. In addition, the implementation of the fourth rods linking the primary structure of the engine pylon to the fixing base makes it possible to reduce the dimensions of the primary structure of the engine pylon, thus allowing and facilitating the fixing of a turbojet engine of greater dimensions to the wing of the aircraft.

(16) Advantageously, said primary structure comprises a top spar, a bottom spar and two starboard and port lateral panels linking the two spars, said fourth fittings being fixed on either side of said primary structure in the bottom part of said lateral panels.

(17) According to a particular aspect, said body comprises a first body part in the form of an inverted U extending inside a second body part in inverted U form, in which the rear part of the primary structure is fixed in the first body part, and in which the two first fixing yokes, the two second fixing yokes and the two fourth fixing yokes are secured to the second body part.

(18) According to another particular aspect, said first body part extends inside said second body part, said first body part and said second body part extending in a same plane, parallel to the vertical plane.

(19) According to yet another particular aspect, said body comprises at least one reinforcing rib linking said first body part and said second body part.

(20) According to a particular aspect, said rear part of the primary structure secured to said fixing base has a height measured in the vertical direction between said top and bottom spars and a width measured in the transverse direction between said lateral panels, and said fixing base has a height measured between the orthogonal projection of an axis of a second link of said second fixing yoke and an axis of a sixth link of said third fixing yoke and a width measured in the transverse direction between said two first fixing yokes, the height of said primary structure being less than the height of said fixing base and the width of said primary structure being less than the width of said fixing base.

(21) According to a particular aspect, said first body part of said fixing base comprises a first port protuberance and a first starboard protuberance extending vertically in a plane that is overall at right angles to the vertical median plane, said first protuberances being disposed on either side of said primary structure in the top part of said lateral panels. For the first port protuberance, the assembly comprises a first link, a first end of which is fixed to said first protuberance by a tenth pivot link. For the first starboard protuberance, the assembly comprises a second link, a first end of which is fixed to said first starboard protuberance by an eleventh pivot link. Said second port and starboard fittings respectively comprise a second port protuberance and a second starboard protuberance extending vertically in a plane that is overall at right angles to the vertical median plane, said second protuberances extending in the bottom part of the second fittings, the second starboard protuberance being fixed to a second end of said first link by an eleventh pivot link and said second port protuberance being fixed to a second end of said second link by an eleventh pivot link.

(22) According to another particular aspect, said tenth and eleventh pivot links have a fitted link axis and at least one of said twelfth and thirteenth pivot links has a link axis comprising a play.

(23) According to a particular aspect, said second, fourth, fifth and sixth links are pivot links or ball joint links.

(24) The invention also proposes an aircraft comprising a propulsion system and an assembly as described previously, in which the propulsion system is fixed to the engine pylon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features of the invention mentioned above, as well as others, will become more clearly apparent on reading the following description of an exemplary embodiment, said description being

given in relation to the attached drawings, in which:
(2) FIG. **1** is a side view of an aircraft according to the invention;
(3) FIG. **2** is a perspective view of an assembly according to an embodiment of the invention;
(4) FIG. **3** is a side view of FIG. **2**; and
(5) FIG. **4** is a rear view partially illustrating the assembly of FIGS. **2** and **3**.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(6) FIG. **1** shows an aircraft **10** which comprises a propulsion system **102**, for example of turbojet or turboprop engine type. The propulsion system **102** is linked to a wing **104** of the aircraft **10** via an engine pylon **106**. The wing **104** and the engine pylon **106** form an assembly **100** according to the invention and the propulsion system **102** is fixed to the engine pylon **106** by any appropriate fixing means known to the person skilled in the art such as those disclosed in the document US-A-2016/0221682.
(7) In the following description, the terms relating to a position are taken with reference to an aircraft in normal flight position, that is to say as is represented in FIG. **1** and the “front” and “rear” positions are taken with respect to the front and the rear of the propulsion system **102** and with respect to the direction of advance F of the aircraft **10** when the propulsion system **102** is operating.
(8) In the following description, and by convention, X denotes the longitudinal direction of the propulsion system which is horizontal when the aircraft is on the ground, Y denotes the transverse direction which is horizontal when the aircraft is on the ground, and Z denotes the vertical direction which is vertical when the aircraft is on the ground, these three directions X, Y and Z being mutually orthogonal.
(9) The engine pylon **106** and the propulsion system **102** have a vertical median plane XZ and the propulsion system **102** is, here, a turboprop engine with a propeller **102***a*, but it could be of the turbofan engine type with a nacelle.
(10) FIG. **2** shows the assembly **100** according to the invention. The engine pylon **106** comprises a rigid structure forming a box and also called primary structure **202**. The primary structure **202** is formed by a top spar **204**, a bottom spar **206** and two starboard and port lateral panels **208** (only the port panel is represented in FIGS. **2** and **3**) linking the two spars **204** and **206**.
(11) The form of the primary structure **202** is, in this example, configured so that the engine pylon **106** represents a front pyramid **201**, situated at the front end of the engine pylon **106** in the direction of advance F, and a base **203**, substantially rectangular, situated at the rear end of the engine pylon **106** in the direction of advance F. The base **203** has a height h measured substantially in the vertical direction Z between the top **204** and bottom **206** spars and a width **1** measured substantially in the transverse direction Y between the two lateral panels **208**.
(12) The structure of the wing **104** comprises a front spar **210**, an upper surface panel **214** and a lower surface panel **216** which are both fixed to the front spar **210** and extend overall in horizontal planes XY.
(13) Obviously, to ensure the rigidity of the wing, its structure comprises other elements such as ribs which are distributed between the upper surface panel **214** and the lower surface panel **216**.
(14) The front spar **210** here takes the form of a profile in the form of a reversed Z-shaped tread in FIGS. **2** and **3**, with a top wing **210***a* (also called top part) and a bottom wing **210***b* (also called bottom part) which are overall horizontal and a central part **210***c* that is overall at right angles to the median plane extending overall vertically parallel to the plane YZ. The upper surface panel **214** is fixed on top of the top wing **210***a* of the spar **210** and the lower surface panel **216** is fixed under the bottom wing **210***b* of the spar **210**. The fixing of the front spar **210** to the panels **214** and **216** is done by any appropriate means such as welds, bolts, et cetera. The upper surface panel **214** is thus fixed in the top part of the front spar **210** and the lower surface panel **216** is fixed in the bottom part of the front spar **210**. The central part **210***c* is level with the leading edge of the wing **104**.
(15) So as to allow the fixing of the engine pylon **106** to the wing **104**, the front spar **210** bears a first fitting **211***a* fixed at the vertical median plane XZ, and more particularly here at the central

part **210**c of the front spar **210**. The lower surface panel **216** bears two second starboard **211**b′ and port **211**b fittings (only the second port fitting is represented in FIGS. **2** and **3**, the second starboard fitting is represented in FIG. **4**) which are fixed onto the outer face of the lower surface panel **216** and in the alignment of the bottom part **210**b of the front spar **210**. In addition, the lower surface panel **216** also bears a third fitting **221** (visible in FIG. **3**), which is, in this example, fixed under the lower surface **216**, substantially at the lowest part of the lower surface **216**. In this example, the third fitting **221** is positioned at the rear of the second starboard **211**b′ and port **211**b fittings.

(16) The assembly **100** further comprises a fixing base **270** secured to the rear part of the primary structure **202**. The fixing base **270** comprises, in this example, a body **270**a extending vertically in the plane YZ. More specifically, the body **270**a has a first body part **272** that takes the form of an inverted U and a second body part **271** that also takes the form of an inverted U. The first body part **272** and the second body part **271** extend in a vertical plane parallel to the vertical plane YZ and the first body part **272** is arranged inside the second body part **271**.

(17) In the example illustrated, the first body part **272** comprises a central top wall **272**a extending horizontally in a plane parallel to the plane XY. The first body part **272** also comprises two starboard **272**b′ and port **272**b lateral walls (only the port lateral wall is represented in FIGS. **2** and **3**, the starboard lateral wall is visible in FIG. **4**) which extend at right angles from the central top wall **272**a and which extend vertically in planes parallel to the vertical median plane XZ. The central top wall **272**a and the starboard **272**b′ and port **272**b lateral walls are secured and extend around and above the rear part of the primary structure **202** and form a housing **273** for receiving the base **203** of the primary structure **202**. The dimensions of the housing **273** are clearly chosen to match the dimensions of the base **203** of the primary structure **202**. It is understood that the primary structure **202** is fixed to the housing **273** by any appropriate means such as welds, bolts, et cetera.

(18) The fixing base **270** comprises at least one reinforcing rib **272**c linking the walls **272**a, **272**b, **272**b′ of the first body part **272** to the second body part **271**. In this example, three ribs **272**c are implemented (only two ribs are visible in FIG. **2**) and extend respectively from the top central wall **272**a and starboard **272**b′ and port **272**b lateral walls to the second body part **271**. These reinforcing ribs **272**c notably allow the fixing base **270** to withstand and transmit the forces originating from the propulsion system **102** to the wing **104**. The ribs **272**c fix together the walls **272**a, **272**b, **272**b′ of the first body part **272** to the walls of the second body part **271**.

(19) So as to allow the engine pylon **106** to be fixed to the wing **104**, the second body part **271** of the body **270**a of the fixing base **270** comprises two first starboard and port fixing yokes **273**a (only the first port fixing yoke is represented in FIGS. **2** and **3**) which extend from the second body part **271** of the body **270**a to the rear and which are disposed in the top part on the sides of the fixing base **270**. Each of the first fixing yokes **273**a is secured to one of the second fittings **211**b, **211**b′ by a first pivot link **283**a, in particular a sliding pivot link.

(20) In this example, the first fixing yokes **273**a and the second fittings **211**b, **211**b′ each comprise a bore into which a shaft is threaded to form a first pivot link **283**a. This first pivot link **283**a has an axis of rotation that is substantially horizontal which extends overall parallel to the transverse direction Y.

(21) The second body part **271** of the body **270**a of the fixing base **270** also comprises two second starboard and port fixing yokes **273**b (only the second port fixing yoke is represented in FIGS. **2** and **3**) which extend from the second body part **271** of the body **270**a to the rear and which are disposed in the bottom part on the sides of the fixing base **270**. The two second fixing yokes **273**b are secured to a shackle **280** which can be triangular, respectively by a first and a second rod **281**a, **281**b.

(22) In this example, the second starboard fixing yoke is fixed to a second end of the first rod **281**a via a second link **283**b, while the second port fixing yoke **273**b is fixed to a second end of the second rod **281**b via another second link **283**b. The second link **283**b has at least one degree of freedom in rotation on an axis extending here substantially parallel to the transverse direction Y. In

this example, the second link **283***b* is a pivot link, the axis of which extends overall parallel to the transverse direction Y.
(23) In a variant that is not illustrated, it would be possible to envisage replacing the second pivot link **283***b* with a ball joint link.
(24) The first ends of the first **281***a* and second **281***b* rods are respectively fixed to the shackle **280** by a third **283***c* and a fourth **283***d* pivot link. The axis of the third **283***c* and fourth **283***d* pivot links extends overall parallel to the direction at right angles to the plane passing through the axes of the first and second rods **281***a*, **281***b*. In this example, the first ends of each of the two first and second rods **281***a*, **281***b* and the shackle **280** comprise bores into which shafts are threaded to form said third **283***c* and fourth **283***d* pivot links.
(25) Furthermore, the shackle **280** is fixed by a fifth link **283***e* to the third fitting **221** of the lower surface **216**. The fifth link **283***e* has at least one degree of freedom in rotation on an axis extending here substantially parallel to the vertical direction Z. In this example, the fifth link **283***e* is a pivot link, the axis of which extends overall parallel to the direction at right angles to the plane passing through the axes of the first and second rods **281***a*, **281***b*. For this, the shackle **280** and the third fitting **221** each have a bore into which a shaft is threaded to form the fifth pivot link **283***e*.
(26) In a variant that is not illustrated, it would be possible to envisage replacing the fifth pivot link **283***e* with a ball joint link.
(27) The second body part **271** of the body **270***a* of the fixing base **270** also comprises at least one third central fixing yoke **273***c* (only one in the example illustrated) which extends from the second body part **271** of the body **270***a* to the rear and which is disposed substantially centrally (that is to say at the vertical median plane XZ) in the top part of the fixing base **270**.
(28) The third central fixing yoke **273***c* is fixed to the first fitting **211***a* by a third rod **282**. More specifically, a first end of the third rod **282** is fixed to the first fitting **211***a* by a seventh link **283***g* and the second end of the third rod **282** is fixed to the third central fixing yoke **273***c* by a sixth link **283***f*. The sixth **283***f* and seventh **283***g* links have at least one degree of freedom in rotation on an axis extending here substantially parallel to the transverse direction Y. In this example, the sixth **283***f* and seventh **283***g* links are pivot links, the axis of which extends overall parallel to the transverse direction Y.
(29) For this, the third rod **282** comprises a bore at each of its ends into each of which a shaft is threaded and cooperates respectively with a bore formed in the third central fixing yoke **273***c* and in the first fitting **211***a* to form the sixth **283***f* and seventh **283***g* pivot links.
(30) In a variant that is not illustrated, it would be possible to envisage replacing the sixth **283***f* and seventh **283***g* pivot links with ball joint links.
(31) According to the invention, the primary structure **202** bears a fourth starboard fitting and a fourth port fitting **301** (only the fourth port fitting is visible in FIGS. **2** and **3**). The fourth fittings **301** are fixed on either side of the primary structure **202**. In this example, the fourth starboard fitting is fixed to the starboard lateral panel and the fourth port fitting **301** is fixed to the port lateral panel **208**. Preferably, and as illustrated in FIGS. **2** and **3**, the fourth fittings **301** are fixed in the lower part of the lateral panels **208** so as to optimize the transmission of the forces originating from the propulsion system **102** and passing through the engine pylon **106** to the wing **104**.
(32) The second body part **271** of the body **270***a* of the fixing base **270** further comprises two fourth starboard and port fixing yokes **302** extending to the front and in the top part on the sides of said fixing base **270**. Each of the fourth fixing yokes **302** is fixed to a second end of a fourth rod **303** by a ninth pivot link **305**. The second end of each fourth rod **303** is fixed to a fourth fitting **301** by an eighth pivot link **304**.
(33) The axis of the eighth **304** and ninth **305** pivot links is overall parallel to the transverse direction Y.
(34) The implementation of the fourth fittings **301**, of the fourth fixing yokes **302** and of the fourth rods **303** makes it possible to reduce the dimensions of the primary structure **202** of the engine

pylon **106** while ensuring an optimized transfer of the forces originating from the propulsion system **102** to the structure of the wing **104** through these additional force paths. Consequently, the reduction of the dimensions of the primary structure **202** of the engine pylon **106** allows and facilitates the installation and the fixing of propulsion systems of dimensions greater than the turbojet engines currently fixed to the wing of the aircraft.
(35) As illustrated in FIG. **4**, which partially represents the assembly according to a rear view, the assembly **100** further comprises a set of two links **275***a*, **275***b* making it possible to react to the forces originating from the propulsion system **102** essentially in the transverse direction Y. In this example, the fixing base **270** comprises, at the first body part **272**, a first port protuberance **274***a* and a first starboard protuberance **274***b*. In this example, each first protuberance **274***a*, **274***b* has an ear form and extends vertically and at right angles to the lateral walls **272***b*, **272***b′*, that is to say in a plane at right angles to the vertical median plane XZ. More specifically, the first port protuberance **274***a* extends substantially in the corner between the central top wall **272***a* and the port lateral panel **208** of the primary structure **202** and the first starboard protuberance **274***b* extends substantially in the corner between the central top wall **272***a* and the starboard lateral panel **208** of the primary structure **202**. The first protuberances **274***a*, **274***b* are disposed in the top part and in the rear part of the fixing base **270**, that is say at the rear of the primary structure **202**.
(36) The second port **211***b* and starboard **211***b′* fittings respectively bear a second port protuberance **274***c* and a second starboard protuberance **274***d*. In this example, each second protuberance **274***c*, **274***d* also has an ear form and extends vertically in a plane at right angles to the vertical median plane XZ. In this example, the second protuberances **274***c*, **274***d* extend in the bottom part of the second fittings **211***b*, **211***b′*.
(37) In this example, the first protuberances **274***a* **274***b* face one another. Likewise, the second protuberances **274***c* and **274***d* also face one another. The set of links **275***a*, **275***b* cross-links the second fittings **211***b*, **211***b′* to the fixing base **270**. More specifically, a first link **275***a* links the first port protuberance **274***a* to the second starboard protuberance **274***d* while the second link **275***b* links the first starboard protuberance **274***b* to the second port protuberance **274***c*. The first **275***a* and second **275***b* links are therefore crossed and extend substantially in two parallel planes at right angles to the vertical median plane XZ. The two parallel planes are slightly spaced apart to favour the crossing of the first **275***a* and second **275***b* links.
(38) The ends of the first link **275***a* are secured to the first port protuberance **274***a* and to the second starboard protuberance **274***d* respectively by a tenth **284***a* and an eleventh **284***b* pivot link. The ends of the second link **275***b* are secured to the first starboard protuberance **274***b* and to the second port protuberance **274***c* respectively by a twelfth **284***c* and a thirteenth **284***d* pivot link.
(39) More specifically, the first link **275***a* is said to be “engaged” whereas the second link **275***b* is said to be “on standby”. That means that the link axes of the tenth **284***a* and eleventh **284***b* pivot links are fitted and therefore react to the forces under stress, such that the first link **275***a* represents the so-called “main” force path for the transmission of the forces having a component that is substantially oriented in the transverse direction Y.
(40) At least one of the twelfth **284***c* and thirteenth **284***d* pivot links has a play which makes it possible, should the first link **275***a* break, to engage the second link **275***b* such that it is the second link **275***b* which reacts to the forces having a component substantially oriented in the transverse direction Y. The second link **275***b* thus represents the so-called “secondary” force path.
(41) For example, the twelfth **284***c* and thirteenth **284***d* pivot links each have a link axis borne respectively by the first starboard protuberance **274***b* and by the second port protuberance **274***c*. At least one of the bores formed at the ends of the second link **275***b* and cooperating with these link axes has, for example, an oblong form such that the second link **275***b* is mounted with a play at at least one of the twelfth **284***c* and thirteenth **284***d* pivot links.
(42) The particular implementation of this set of links **275***a*, **275***b* also makes it possible to reduce the dimensions of the primary structure **202** of the engine pylon **106** while ensuring an optimized

transfer of the forces originating from the propulsion system **102** to the structure of the wing **104**.
(43) The fixing base **270** has a height H which is measured between the orthogonal projection of the axis of the second fixing yoke **273***b* and the axis of the third fixing yoke **273***c* (that is to say vertically in the vertical direction Z) while the width L is measured horizontally (that is to say in the transverse direction Y) between the two first fixing yokes **273***a*.
(44) Preferably, and in order to compensate for the lower inertia of the assembly **100**, the height h of the primary structure **202** is less than the height H of the fixing base **270** and the width **1** of the primary structure **202** is less than the width L of the fixing base **270**.
(45) This way, the draining of the forces passing through the primary structure **202** to the fixing base **270** and the associated rods/shackles is optimized. Indeed, the implementation of a width L of the fixing base **270** greater than the width **1** of the primary structure **202** favours the absorption of the moment about the vertical axis Z. Likewise, with the height H of the fixing base **270** being greater than the height h of the primary structure **202**, the points governing the moments about the lateral axis Y are further apart. These two conditions make it possible to optimize the lever arms engaged in the absorption of the moments. This disposition therefore ensures, locally, an inertia greater than that of the primary structure **202**, thus making it possible to limit the dimensions of the primary structure **202**.
(46) It is thus possible to propose a primary structure **202** that has a lesser bulk than the primary structures of the prior art solutions.
(47) The primary structure **202** does however have standard dimensions at the front pyramid **201** of the primary structure **202** so as to ensure the compatibility of the assembly **100** to the fixing of propulsion systems currently implemented.
(48) While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

1. An assembly for mounting, on an aircraft, a propulsion system having a vertical median plane, said assembly comprising: a wing with a front spar, an upper surface panel fixed in a top part of said front spar and a lower surface panel fixed in a bottom part of said front spar, wherein the front spar bears a first fitting at a vertical median plane, wherein said lower surface panel bears two second fittings, a second starboard fitting and a second port fitting, at the bottom part of said front spar, and wherein the lower surface panel bears a third fitting positioned at a rear of said second starboard and port fittings; an engine pylon comprising a primary structure; a shackle fixed by a fifth link to the third fitting of said lower surface panel, said fifth link having at least one degree of freedom in rotation; a first rod and a second rod, wherein a first end of each first rod and second rod is fixed to the shackle respectively by a third pivot link and a fourth pivot link; a third rod, wherein a first end of the third rod is fixed to the first fitting by a seventh link, said seventh link having at least one degree of freedom in rotation; a fixing base comprising a body secured to a rear part of the primary structure, said body comprising two first fixing yokes, a first starboard fixing yoke and a first port fixing yoke, extending to a rear and in a top part on sides of said fixing base, each of said first fixing yokes being fixed to one of said two second fittings by a first pivot link,

said body further comprising two second fixing yokes, a second starboard fixing yoke and a second port fixing yoke, extending to the rear and in a bottom part on the sides of said fixing base, each of said two second fixing yokes being fixed by a second link to a second end respectively of the first and of the second rods, said second link having at least one degree of freedom in rotation, said body comprising at least one third central fixing yoke extending to the rear and at the vertical median plane, said at least one third central fixing yoke being fixed to a second end of said third rod by a sixth link, said sixth link having at least one degree of freedom in rotation; a fourth starboard fitting and a fourth port fitting, each of said fourth fittings being fixed to the primary structure of said engine pylon; and, two fourth rods, wherein a first end of each fourth rod is fixed to said respective fourth fitting by an eighth pivot link, wherein said body further comprises two fourth fixing yokes, a fourth starboard fixing yoke and a fourth port fixing yoke, extending to a front and in the bottom part on the sides of said fixing base, each of said fourth fixing yokes being fixed to a second end of one of the two fourth rods by a ninth pivot link, and, wherein said body comprises a first body part in a form of an inverted U extending inside a second body part having an inverted U form as well, wherein the rear part of the primary structure is fixed in the first body part, and wherein the two first fixing yokes, the two second fixing yokes, and the two fourth fixing yokes are secured to the second body part.

2. The assembly according to claim 1, wherein said primary structure comprises a top spar, a bottom spar, and two lateral panels, a starboard lateral panel and a port lateral panel, linking the top and bottom spars, said fourth fittings being fixed on either side of said primary structure in a bottom part of said lateral panels.

3. The assembly according to claim 1, wherein said body comprises at least one reinforcing rib linking said first body part and said second body part.

4. The assembly according to claim 2, wherein said rear part of the primary structure secured to said fixing base has a height (h) measured in a vertical direction (Z) between said top and bottom spars and a width (**1**) measured in a transverse direction (Y) between said lateral panels, and wherein said fixing base has a height (H) measured between an orthogonal projection of an axis of the second link of said second fixing yoke and an axis of the sixth link of said third fixing yoke and a width (L) measured in the transverse direction (Y) between said two first fixing yokes, the height (h) of said primary structure being less than the height (H) of said fixing base and the width (**1**) of said primary structure being less than the width (L) of said fixing base.

5. The assembly according to claim 2, wherein said first body part of said fixing base comprises a first port protuberance and a first starboard protuberance extending vertically in a plane that is at right angle to the vertical median plane, said first port and starboard protuberances being disposed on either side of said primary structure in a top part of said lateral panels, wherein, for the first port protuberance, the assembly comprises a fifth rod, a first end of which is fixed to said first port protuberance by a tenth pivot link, wherein, for the first starboard protuberance, the assembly comprises a sixth rod, a first end of which is fixed to said first starboard protuberance by an eleventh pivot link, and wherein said second port and starboard fittings respectively comprise a second port protuberance and a second starboard protuberance extending vertically in a plane that is at right angle to the vertical median plane, said second port and starboard protuberances extending in a bottom part of the second fittings, the second starboard protuberance fixed to a second end of said fifth rod by a twelfth pivot link, and said second port protuberance fixed to a second end of said sixth rod by a thirteenth pivot link.

6. The assembly according to claim 5, wherein said tenth and eleventh pivot links each has a fixed link axis, and wherein at least one of said twelfth and thirteenth pivot links has a play.

7. The assembly according to claim 1, wherein said second, fifth, sixth and seventh links are pivot links or ball joint links.

8. An aircraft comprising: a propulsion system, and the assembly according to claim 1, wherein the propulsion system is fixed to the engine pylon.